

A disproof of Conjecture 00000002251

The kernel of the difference operator on meromorphic functions is not one-dimensional

1 The conjecture

Conjecture 00000002251 reads (English version):

Definition: The difference operator $\Delta_c f = f(z+c) - f(z)$. *Conjecture:* The kernel of Δ_c has dimension 1, and the spectrum of Δ_c on meromorphic function spaces (differences of difference polynomials) is explicit of the $\{2 \sin(kc/2)\}$ type (difference spectrum).

The Chinese version says the same: the kernel of Δ_c has dimension 1, and the spectrum of Δ_c on the meromorphic function space (*ya chun han shu kong jian*) is explicitly of $\{2 \sin(kc/2)\}$ type. The statement is a conjunction. Its first clause (kernel of dimension 1) is unhedged; the second clause is hedged by the word “type”. We refute the first clause, which refutes the conjunction, and we also compute the spectrum exactly.

2 Reading and conventions

Definition 1. $\mathcal{M}(\mathbb{C})$ is the set of meromorphic functions $f : \mathbb{C} \rightarrow \mathbb{C}$ (meromorphic at every point of $\mathbb{C}$), a complex vector space under pointwise operations. For $c \in \mathbb{C}$, $\Delta_c : \mathcal{M}(\mathbb{C}) \rightarrow \mathcal{M}(\mathbb{C})$ is $(\Delta_c f)(z) = f(z+c) - f(z)$; it maps $\mathcal{M}(\mathbb{C})$ into $\mathcal{M}(\mathbb{C})$ because translates, sums and scalar multiples of meromorphic functions are meromorphic, and it is $\mathbb{C}$ -linear.

- **Domain.** We read Δ_c on the space the text names, “meromorphic function spaces”, with no growth, periodicity or polynomial restriction (Section 6 lists restricted readings, which we do *not* refute).
- **Values at poles.** Following Mathlib, a meromorphic function is an honest function $\mathbb{C} \rightarrow \mathbb{C}$ with some value at each pole. Lemma 4 shows that the witnesses stay independent if functions that agree off a discrete set are identified, so the result does not depend on this convention.
- **Dimension** is the rank of a vector space (a cardinal, `Module.rank`); “dimension 1” means rank = 1. **Spectrum** is the algebraic spectrum in $\text{End}_{\mathbb{C}}(\mathcal{M}(\mathbb{C}))$: $\mu \in \sigma(\Delta_c)$ iff $\mu \cdot \text{id} - \Delta_c$ is not bijective.
- **The shift** c ranges over all of $\mathbb{C}$; the kernel result holds for every c , including the degenerate $c = 0$. The spectrum result is stated for $c \neq 0$.

3 The kernel has dimension at least two

Lemma 2. For $b, c \in \mathbb{C}$ let $e_b(z) = e^{bz}$. Then $\Delta_c e_b = (e^{bc} - 1) e_b$.

Proof. $e^{b(z+c)} - e^{bz} = e^{bz} e^{bc} - e^{bz}$. □

Theorem 3. *For every $c \in \mathbb{C}$, $\text{rank}_{\mathbb{C}} \ker \Delta_c > 1$. In particular $\ker \Delta_c$ never has dimension 1.*

Proof. Case $c \neq 0$. Let $\omega = 2\pi i/c$, $u = 1$ and $v = e_\omega$. Then $\Delta_c u = 0$ and, by Lemma 2, $\Delta_c v = (e^{2\pi i} - 1)v = 0$. If $su + tv = s'u + t'v$, evaluate at $z = 0$ and at $z = c/2$, where $v(0) = 1$ and $v(c/2) = e^{\pi i} = -1$: $s + t = s' + t'$ and $s - t = s' - t'$, hence $s = s'$, $t = t'$. So u, v are linearly independent elements of $\ker \Delta_c$.

Case $c = 0$. $\Delta_0 = 0$, so $u = 1$ and $v(z) = z$ lie in the kernel; evaluating at $z = 0$ and $z = 1$ gives independence as above.

In both cases $\ker \Delta_c$ contains a linearly independent pair, so its rank is at least $2 > 1$. $\square$

Lemma 4. *Let $c \neq 0$ and $s, t \in \mathbb{C}$. If $s + te^{2\pi iz/c} = 0$ for all z outside a discrete subset of $\mathbb{C}$ (i.e. on a set in the codiscrete filter), then $s = t = 0$.*

Proof. The function $F(z) = s + te^{2\pi iz/c}$ is continuous and vanishes on a punctured neighbourhood of every point, hence everywhere. Now use $z = 0$ and $z = c/2$ as before. $\square$

Remark 5. Both witnesses for $c \neq 0$ are entire, and $|e^{bz}| = e^{\text{Re}(bz)} \leq e^{|b||z|}$. Hence the same pair lies in the kernel of Δ_c on entire functions and on any class of meromorphic functions that contains entire functions of exponential type, so those readings are refuted as well.

4 The spectrum

Theorem 6. *Let $c \neq 0$. Every $\mu \neq -1$ is an eigenvalue of Δ_c on $\mathcal{M}(\mathbb{C})$, and $-1 \notin \sigma(\Delta_c)$. Hence $\sigma(\Delta_c) = \mathbb{C} \setminus \{-1\}$, which is uncountable; in particular $\sigma(\Delta_c) \neq \{2 \sin(kc/2) : k \in \mathbb{Z}\}$, and it is not the image of any countable index set.*

Proof. For $\mu \neq -1$ put $b = \log(1 + \mu)/c$ (any branch; $1 + \mu \neq 0$). Then $e^{bc} = 1 + \mu$, and by Lemma 2 $\Delta_c e_b = \mu e_b$ with $e_b(0) = 1 \neq 0$. For $\mu = -1$: $(-\text{id} - \Delta_c)f(z) = -f(z + c)$, which is a bijection of $\mathcal{M}(\mathbb{C})$ with inverse $g \mapsto -g(\cdot - c)$. Finally $\mathbb{C} = (\mathbb{C} \setminus \{-1\}) \cup \{-1\}$ is uncountable, while the image of a countable set is countable. $\square$

Note that $0 \in \sigma(\Delta_c)$: it is the eigenvalue of the (non-trivial) kernel.

5 Relation to the earlier PR #270

PR #270 (orionsheep, merged 2026-10-03) used the same reading and the same witness $e^{2\pi iz/c}$. It was removed in the re-audit commit 541cf4fb, whose note reads “#2251 (4-point cyclic shadow; meromorphic kernel prose-only)”. Its Lean file defined a function `unit4 : Nat -> Int x Int` cycling through four pairs labelled $1, i, -1, -i$ and proved `unit4 (z+4) = unit4 z` by `rfl`, plus the distinctness of four integer pairs. No complex function, no meromorphic function, no operator Δ_c and no kernel or dimension appeared in Lean; the theorem that the kernel has dimension ≥ 2 was prose only, so the Lean certified nothing about the conjecture’s objects.

PR #270 also stated that the eigenvalues $e^{\lambda c} - 1$ of the eigenfunctions $e^{\lambda z}$ “sweep $\mathbb{C} \setminus \{0\}$ ”. That is wrong: as λ ranges over $\mathbb{C}$ (and $c \neq 0$), $e^{\lambda c}$ takes every nonzero value, so $e^{\lambda c} - 1$ takes exactly the values in $\mathbb{C} \setminus \{-1\}$. The value 0 is attained (it is the eigenvalue of the non-trivial kernel, contradicting PR #270’s own kernel claim), and -1 is the omitted value (Theorem 6). The present submission corrects both defects: Δ_c is a $\mathbb{C}$ -linear endomorphism of the genuine space of meromorphic functions $\mathbb{C} \rightarrow \mathbb{C}$, and the rank of its kernel and its spectrum are theorems about that object.

6 Readings that are not refuted

The kernel clause can hold if Δ_c is restricted to a smaller space that excludes non-constant periodic functions. We do *not* refute such readings, for example: Δ_c on polynomials or rational functions (for $c \neq 0$ a c -periodic one is constant: if z_0 is not a pole, then $f - f(z_0)$ vanishes at $z_0 + nc$ for all $n \in \mathbb{N}$); on meromorphic functions of order < 1 ; or on 2π -periodic trigonometric polynomials $\sum a_k e^{ikz}$ with $c/2\pi$ irrational. In the last setting $\Delta_c e^{ikz} = (e^{ikc} - 1)e^{ikz} = 2i e^{ikc/2} \sin(kc/2) e^{ikz}$, which is presumably where the $\{2 \sin(kc/2)\}$ set comes from.

We take the literal reading because the text itself names the domain: “the spectrum of Δ_c on meromorphic function spaces”, with “difference polynomials” (polynomial expressions in shifts $f(z + c_j)$) as the objects Δ_c acts on. No growth bound, periodicity or polynomial restriction is stated, and the kernel clause carries no hedge. On that space the kernel of Δ_c is the space of all c -periodic meromorphic functions, which has dimension at least 2 for every c .

7 Lean 4 formalization

The file `lean/Conjecture2251/Basic.lean` (221 lines, only `import Mathlib`; Lean v4.33.1, Mathlib v4.33.1) defines

- `MeroFun` : `Submodule C (C -> C)`, carrier $\{f \mid \text{Meromorphic } f\}$ (Mathlib’s Meromorphic: meromorphic at every point);
- `diffOp c` : `Module.End C MeroFun`, $f \mapsto \text{fwdDiff } c \ f$, i.e. $z \mapsto f(z + c) - f(z)$ (Mathlib’s forward difference), with closure from `Meromorphic.meromorphic_fun_comp_add_const_iff_meromorphic`.

Theorems (all in namespace `Tlmc2251`):

- `diffOp_expMF`: $\Delta_c e_b = (e^{bc} - 1)e_b$ (Lemma 2); `expMF_differentiable`, `expMF_growth`: e_b is entire and $\|e^{bz}\| \leq e^{\|b\|\|z\|}$;
- `kernel_pair`: for every c , two elements of $\ker \Delta_c$ with $su + tv = s'u + t'v \Rightarrow s = s', t = t'$;
- `rank_ker_diffOp_gt_one`: $1 < \text{Module.rank } C (\text{LinearMap.ker } (\text{diffOp } c))$ for all c (Theorem 3); `kernel_not_dim_one`: the rank is $\neq 1$;
- `indep_mod_codiscrete`: Lemma 4, with the filter `codiscrete C`;
- `hasEigenvalue_diffOp`, `neg_one_not_mem_spectrum`, `spectrum_diffOp` (`spectrum C (diffOp c) = \{-1\}^c` for $c \neq 0$), `spectrum_not_countable`, `spectrum_ne_sine_set` (Theorem 6);
- `conjecture_2251_refuted`: the conjunction of `rank_ker_diffOp_gt_one` for all c and, for $c \neq 0$, `spectrum_diffOp` and `spectrum_not_countable`.

Here C stands for $\mathbb{C}$. `lake build` succeeds, and `#print axioms` for `conjecture_2251_refuted`, `kernel_not_dim_one`, `indep_mod_codiscrete`, `spectrum_ne_sine_set` and `expMF_growth` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry`, no `native_decide` and no added axiom.